generated_report_2
tcga_brca: objective-aware integration

> **Integration verdict: integrate_with_caution**
>
> The validation-selected representation is fusion_late (supervised_fusion). Its selected encoders are Frozen UNI2-h tile encoder, gated attention MIL image branch, clinical MLP with train-fitted tabular preprocessing, with late fusion and cross_entropy objective. Held-out comparisons and uncertainty appear below.

Objective and principal findings

Learn a patient-level joint representation of diagnostic H&E whole-slide images and baseline clinical variables that predicts PAM50 intrinsic subtype on held-out patients, quantifies whether each modality adds information, and assesses exploratory overall-survival stratification and subtype-consistent retrieval. Training labels may be used.

Allowed supervision: **train_labels**. Primary metrics: macro_f1, balanced_accuracy. The deterministic planner and templated reporting used no external LLM.

Late-fusion training averages the image and clinical branches' class logits. The exported joint representation concatenates their learned latent vectors; the common logistic probe below is fitted on that concatenation. The MIL image branch pools frozen tile features, while the clinical branch encodes the allowed table.

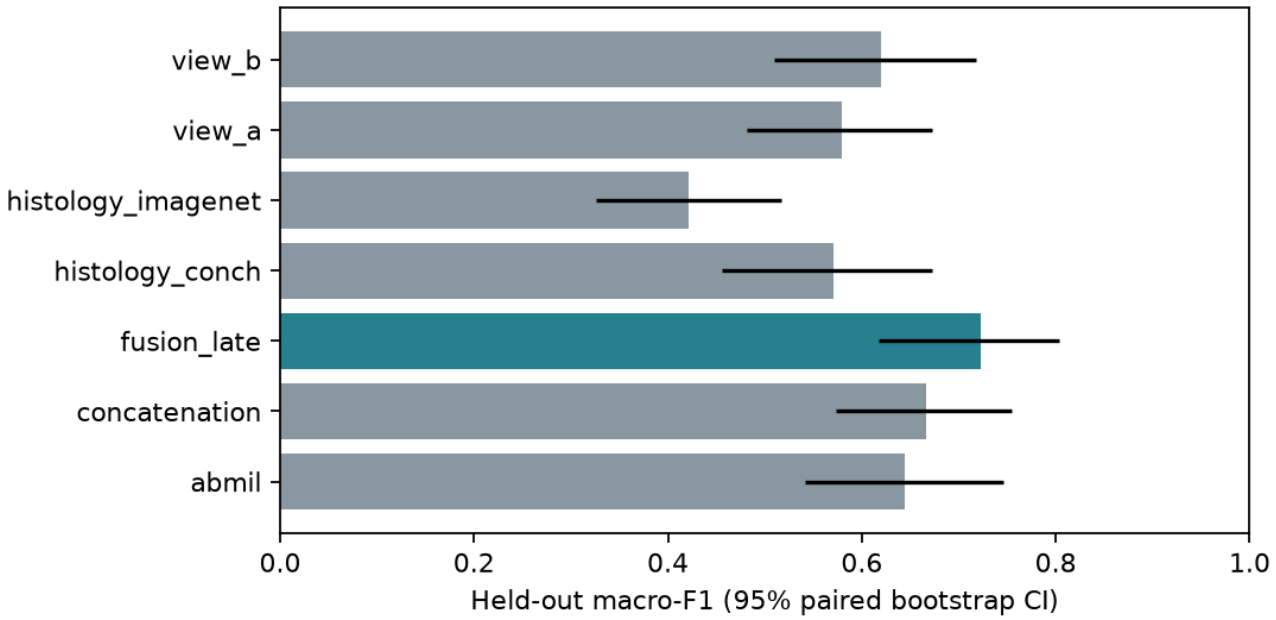
Representation	Family	Dim.	Macro-F1	95% CI	Balanced accuracy
abmil	pretrained_foundation	32	0.645	0.542–0.746	0.644
concatenation	baseline_fusion	1569	0.666	0.574–0.755	0.667
fusion_late	supervised_fusion	64	0.723	0.618–0.804	0.714
histology_conch	baseline_unimodal	768	0.570	0.456–0.672	0.557
histology_imagenet	baseline_unimodal	512	0.421	0.326–0.517	0.419
view_a	baseline_unimodal	1536	0.579	0.481–0.673	0.579
view_b	baseline_unimodal	33	0.620	0.509–0.718	0.615

All rows use the same frozen-embedding logistic probe protocol. End-to-end head predictions are a separate analysis. The selected model was frozen before test access. Paired bootstrap resampling follows the sample unit recorded in the evidence.

The separately evaluated end-to-end head has macro-F1 0.482 and balanced accuracy 0.499. The primary comparisons and confidence intervals use the common logistic probe.

Baseline names: view_a uses histology; view_b uses clinical. Concatenation uses the scaled feature blocks from both modalities.

Against the best single modality (abmil), the selected-minus-reference macro-F1 difference is 0.078 (95% CI -0.009 to 0.170). Project decision thresholds are pre-registered in METHOD_NOTES.md and are not external scientific standards.



Held-out comparison; candidate selection used validation only.

Data, pairing and cohort

Modality	Type	Samples	Features	Missingness/QC
histology	wsi_image	1061	0	see modality-specific QC and clinical missingness artifacts
clinical	tabular_clinical	1061	6	see modality-specific QC and clinical missingness artifacts

Pairing level: patient. All eligible samples were assigned before fitting. Retrospective clinical data; missing covariates. Histology may encode tissue source site. PAM50 is an RNA-derived target; no RNA-derived input fields allowed.

Cohort step	Count
all clinical deaths	149
all clinical os eligible	1081
complete clinical	872
excluded slides	1
labeled and complete	665
labeled four class	802
manifest slides	1133
normal like excluded primary	23
primary slide patients	1061
primary slides	1132
with pam50 including normal	825

Preprocessing and leakage control

Fit scope: training only. Preprocessing receipts record fitted identifiers, split hashes, QC thresholds and feature selection. Neither test data nor target labels fit preprocessing.

Clinical inputs: age, stage, histological_type, er, pr, her2. train median numeric; missing category categorical. Missingness indicators: all original variables. Molecular-derived, outcome, follow-up and treatment variables are excluded from both tabular and templated-text inputs.

WSI preprocessing uses per-slide saturation-Otsu segmentation with morphology, followed by physical-coordinate sampling and pixel-level tissue/blur/pen QC. 1132 of 1132 slides produced coordinate stores. Requested fields: 224 pixels at 0.500 µm/px; the same physical field is resized to each encoder's native input. Stain normalization is disabled.

After a region-QC failure, 1 slide received an explicitly logged coordinate repair: large, nearly uniform, highly saturated edge artifacts were excluded from the segmentation thumbnail before Otsu thresholding. The raw slide, original coordinates and failure evidence were preserved. Region-QC thresholds and the production tile cap stayed fixed; no patient was excluded. The evidence bundle records original/repaired coordinate hashes and the postprocessor source.

The measured pilot used 20 training slides. The fixed cap is 128 tiles/slide, with safety factor 2, projected extraction cost 3.941 aggregate GPU-hours against the 8.000-hour envelope.

Before TCGA fitting, the operational queue limit changed the budget-only candidate cap 256 to the selected cap above. Partition h100_short allows 2.000 hours/task; the feature array runs at most 2 shard at a time. The policy retains measured twofold safety plus 90 seconds of initialization allowance per task. All methods use the same production cap.

Frozen tile encoder	Measured encode tiles/second
CONCH-Z	443.869
UNI2-h	337.142
imagenet_resnet18	1587.785

Agent choices and empirical pilot

Candidate	Status	Rule and reason
view_a	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
view_b	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
concatenation	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
wnn	not applicable	COUNT_LIKELIHOOD_OR_NEIGHBOR_INPUT: Requires paired RNA/ATAC count modalities
multivi	not applicable	COUNT_LIKELIHOOD_OR_NEIGHBOR_INPUT: Requires paired RNA/ATAC count modalities
mofa	not applicable	RAW_VIEW_FACTOR_MODEL: Gigapixel images require a patch/MIL encoder; direct numeric factor model is not a raw-image encoder
simclr	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
clip_dual	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective; small-pair regime: contrastive gains must be established empirically
fusion_early	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective

Candidate	Status	Rule and reason
fusion_late	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
fusion_gated	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
abmil	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
fm_last_block	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
fm_image_text	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective; small-pair regime: contrastive gains must be established empirically
histology_conch	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
histology_imagenet	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective

Selection rule: EPSILON_PARETO_SEQUENTIAL_QUALITY_CONSTRAINTS_THEN_SIMPLICITY. Validation shortlist: fusion_late. Separate evidence columns preserve utility, alignment, structure, validity, stability and compute rather than an unexplained weighted score.

Pilot caps: 600 adapter-fit seconds and 50 epochs/iterations; declared seeds: 774, 775, 776. Every eligible candidate is attempted. Import and data-loading time are excluded from adapter fit time. Shared PCA/LSI fitting occurs in preprocessing; this is not a claim of zero total baseline cost.

Pilot candidate	Successful / attempted	Validation macro-F1 (seed SD)	Recall@10	Failure evidence
abmil	3 / 3	0.696 (0.012)	Not applicable	None
clip_dual	3 / 3	0.711 (0.031)	0.337	None
concatenation	1 / 1	0.720	0.322	None
fm_image_text	3 / 3	0.453 (0.006)	0.145	None
fm_last_block	3 / 3	0.593 (0.026)	0.054	None
fusion_early	3 / 3	0.729 (0.020)	0.061	None
fusion_gated	3 / 3	0.723 (0.014)	0.125	None
fusion_late	3 / 3	0.759 (0.012)	0.054	None
histology_conch	1 / 1	0.605	0.287	None
histology_imagenet	1 / 1	0.535	0.284	None
simclr	3 / 3	0.692 (0.010)	0.274	None
view_a	1 / 1	0.665	0.322	None
view_b	1 / 1	0.568	0.322	None

Actual tower fine-tuning is capped at 128 training patients with four cached tiles per patient; training samples one cached tile per patient step and exported inference uses the first cached tile. This differs from full-cohort frozen-feature MIL and is recorded as unequal data/capacity access. Clinical templated text adds no information beyond the allowed table.

Primary-metric differences after freezing

Metric	Selected minus best single (95% CI)	Selected minus simple fusion (95% CI)
balanced_accuracy	0.070 (-0.021, 0.152)	0.047 (-0.033, 0.119)
macro_f1	0.078 (-0.009, 0.170)	0.056 (-0.035, 0.141)

Model	Seed	Fit seconds	Epochs/steps	Stopping
view_a	774	0.0	0	family-specific convergence/max iterations
view_b	774	0.0	0	family-specific convergence/max iterations
concatenation	774	0.0	0	family-specific convergence/max iterations
abmil	774	5.5	17	validation early stopping
abmil	775	5.4	20	validation early stopping
abmil	776	4.8	16	validation early stopping
histology_conch	774	0.0	0	family-specific convergence/max iterations
histology_imagenet	774	0.0	0	family-specific convergence/max iterations
fusion_late	774	5.6	18	validation early stopping
fusion_late	775	5.4	20	validation early stopping
fusion_late	776	6.3	21	validation early stopping

Project aggregate GPU time in the saved ledger: 2.770 GPU-hours (reconciled 2026-11-03T11:00:33.473861+00:00). This includes both datasets and runtime/throughput acceptance work, rather than only this dataset. The immutable ledger snapshot accompanies this report.

Representation quality and confounding

Model	Effective rank	Leiden ARI	Leiden NMI	Alignment
abmil	6.28	0.117	0.266	Not applicable
concatenation	115.37	0.099	0.238	FOSCTTM 0.190; pool 160
fusion_late	8.40	0.119	0.283	FOSCTTM 0.436; pool 160
histology_conch	56.90	0.026	0.096	FOSCTTM 0.210; pool 160
histology_imagenet	86.57	-0.006	0.069	FOSCTTM 0.253; pool 160
view_a	125.93	-0.009	0.063	FOSCTTM 0.190; pool 160
view_b	17.59	0.099	0.263	FOSCTTM 0.190; pool 160

Graph connectivity, global-distance correlations, each modality's neighborhood preservation, per-class results, QC/depth or site predictability, seed/subsample stability and modality ablations are saved as distinct evidence categories.

Alignment definitions from the frozen run records: concatenation, histology_conch, histology_imagenet, view_a, view_b: train-fitted ridge cross-view alignment probe. fusion_late: raw late-fusion tower coordinates; independent classifiers and no embedding-alignment loss. A ridge alignment probe uses training pairs to map the first embedding into the second embedding's coordinates; its retrieval scores describe that inductive cross-view probe. They do not establish that independently learned or single-modality embeddings intrinsically share a coordinate system.

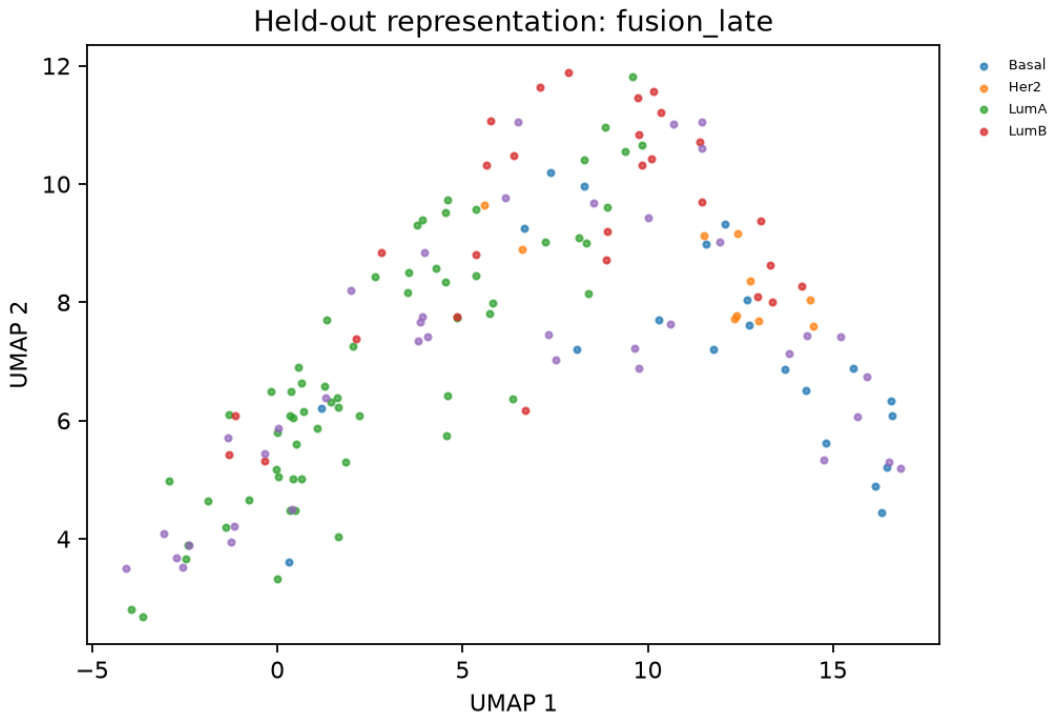
The selected late-fusion towers have independent classifiers and no embedding-alignment loss. Their raw cosine retrieval/FOSCTTM values describe this fitted parameterization and do not identify a semantically aligned cross-modal space. The integration decision uses the predeclared predictive metrics and modality-contribution evidence.

Selected-model class	Support	Recall (95% CI)	OVR AUROC (95% CI where evaluated)	Graph connectivity
Basal	21	0.762 (0.571, 0.905)	0.962 (0.904, 0.996)	0.905
Her2	10	0.700 (0.400, 0.952)	0.967 (0.927, 0.997)	0.800
LumA	62	0.823 (0.726, 0.912)	0.896 (0.835, 0.943)	1.000
LumB	28	0.571 (0.393, 0.750)	0.826 (0.732, 0.910)	0.786

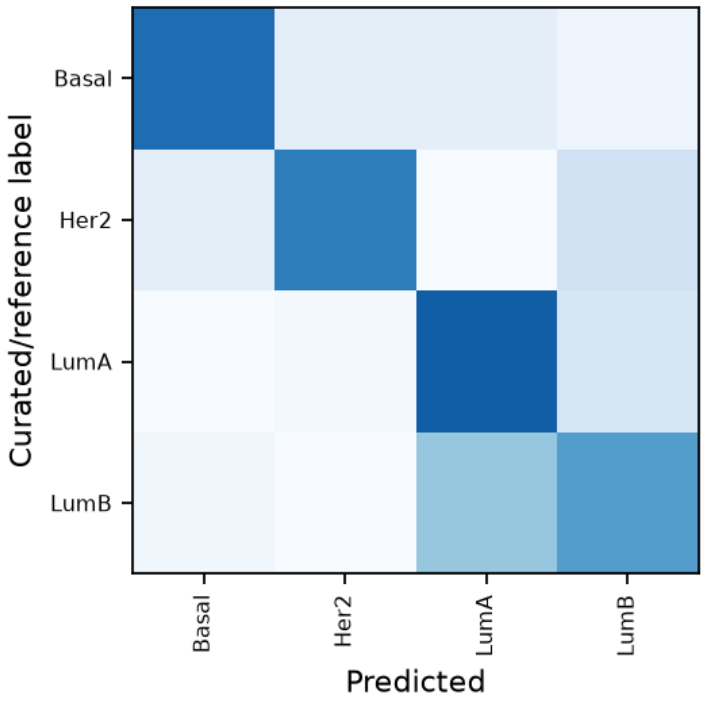
These intervals resample the observed test units. A rare class with only correct observed predictions can have a degenerate recall interval; this does not establish perfect population recall. Interpret class-level estimates together with their support.

Confounding diagnostic	Held-out score	Reference
site	Site balanced accuracy 0.309	Chance 0.027; uniform over training sites

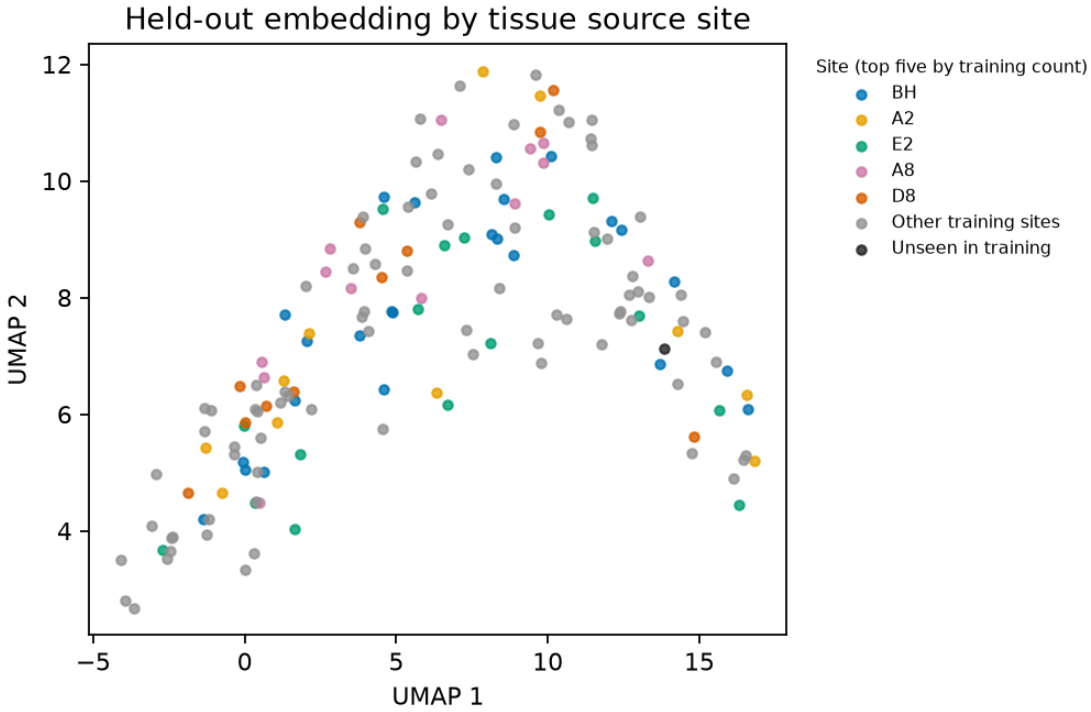
Validity/confounding cautions: Selected representation predicts tissue source site above the project caution margin



selected umap. UMAP is a train-fitted visualization and is not selection evidence.



confusion matrix. Held-out predictions from the frozen model; colors are row-normalized reference-class proportions.



site umap. Sites are grouped using the five largest training-site counts, with other training sites pooled and previously unseen sites separate. UMAP fits training embeddings and transforms test; display only.

Downstream analyses and sensitivity

Executed downstream tasks: classification, retrieval, survival. Full machine-readable results accompany this report.

Exploratory survival status: success. Held-out C-index 0.665; 95% CI 0.558 to 0.770; training events 97, held-out events 30. Retrospective exploratory survival analysis with few held-out events; no clinical claim.

Sensitivity analyses: baseline_subsample_stability; luminal_a_vs_b; seed_stability; site_held_out (success); subsample_stability; without_receptors (success); . Read their separate metrics and fit scopes in downstream/results.json; unresolved analyses remain explicit.

Cross-modal alignment in the fixed held-out candidate pool

Model	A→B recall@1	A→B recall@5	A→B recall@10	B→A recall@10	Pool
concatenation	0.081	0.219	0.356	0.344	160
fusion_late	0.013	0.056	0.081	0.106	160
histology_conch	0.056	0.225	0.325	0.281	160
histology_imagenet	0.037	0.144	0.244	0.256	160
view_a	0.081	0.219	0.356	0.344	160
view_b	0.081	0.219	0.356	0.344	160

Held-out masking with the frozen probe

Model	Masked modality index	Macro-F1	Balanced accuracy
abmil	0	0.038	0.250
abmil	1	0.645	0.644
fusion_late	0	0.307	0.433
fusion_late	1	0.584	0.657

Selected representation across the predeclared seeds

Seed	Validation macro-F1	Test macro-F1	Test balanced accuracy
774	0.746	0.723	0.714
775	0.774	0.748	0.749
776	0.757	0.742	0.744

Training-subsample stability, conditional on fixed preprocessing

Model	Seed	Training fraction	Validation neighbor overlap
fusion_late	774	0.800	0.674
fusion_late	775	0.800	0.622
fusion_late	776	0.800	0.661
concatenation	774	0.800	0.990
concatenation	775	0.800	0.991
concatenation	776	0.800	0.987
view_a	774	0.800	0.997
view_a	775	0.800	0.997
view_a	776	0.800	0.994
view_b	774	0.800	0.980

Model	Seed	Training fraction	Validation neighbor overlap
view_b	775	0.800	0.988
view_b	776	0.800	0.985

Refitted sensitivity analyses

Analysis	Macro-F1	Balanced accuracy
site held out	0.679	0.654
without receptors	0.583	0.563

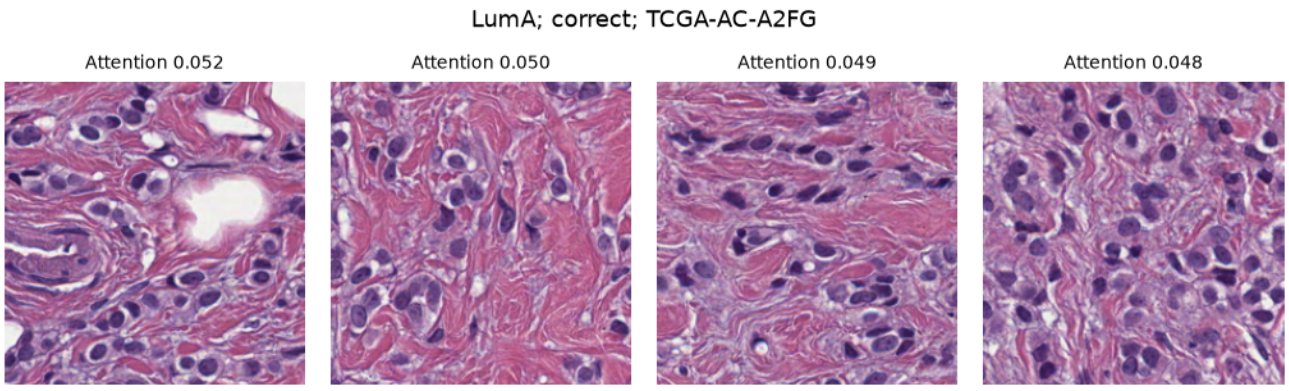
LumA/LumB diagnostic from the frozen four-class confusion matrices

Model	Luminal test patients	Luminal macro-F1
abmil	90	0.682
concatenation	90	0.702
fusion_late	90	0.708
histology_conch	90	0.760
histology_imagenet	90	0.649
view_a	90	0.665
view_b	90	0.537

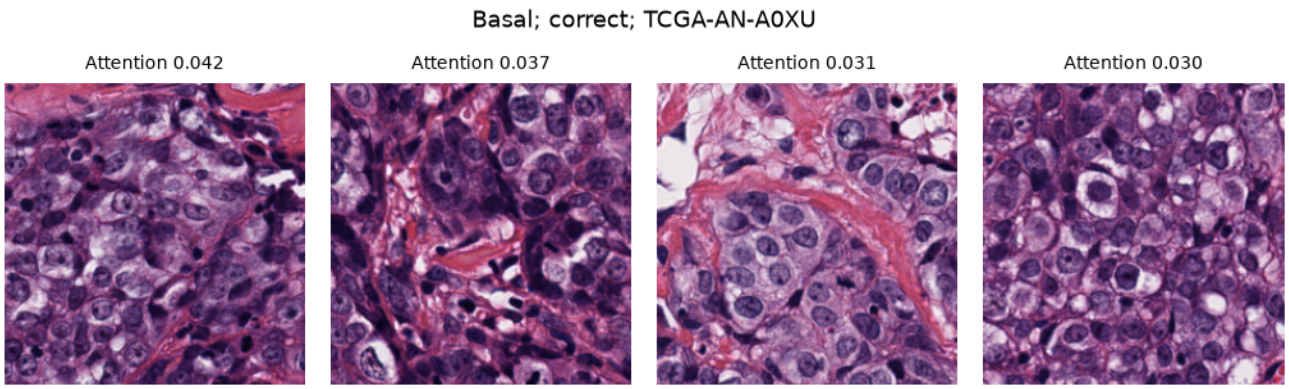
Patient/cell neighbors in the joint representation

Candidate pool	Label agreement@5	Label agreement@10
160	0.463	0.451

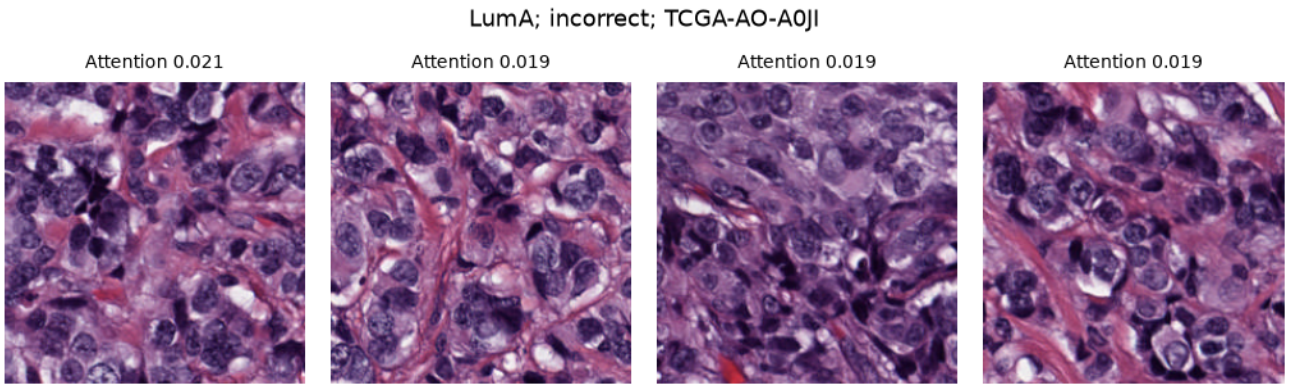
Alignment is descriptive where per-modality spaces are defined; baseline projections are reference spaces, not learned contrastive towers. Masking index 0 denotes the first configured modality and index 1 the second. Seed rows measure training randomness; neighbor overlap measures adapter refits on 80% training samples while retaining fixed train-only preprocessing. Nearest-neighbor label agreement uses labeled queries in the declared pool; candidates without a reference label count as nonmatches. Luminal diagnostics, when present, restrict true labels to LumA/LumB and retain the frozen four-class predictions, including errors into other classes. They are descriptive subgroup comparisons without separate model selection or subgroup confidence intervals.



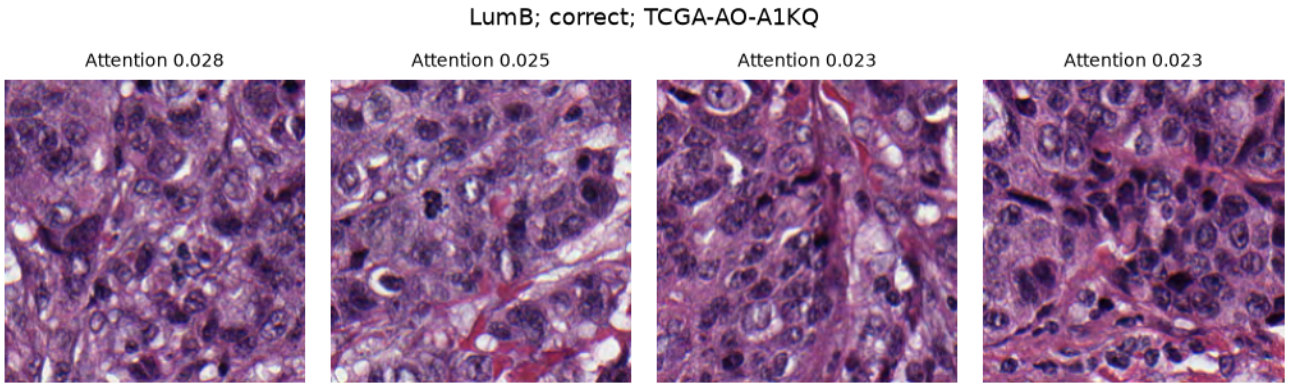
attention TCGA-AC-A2FG. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



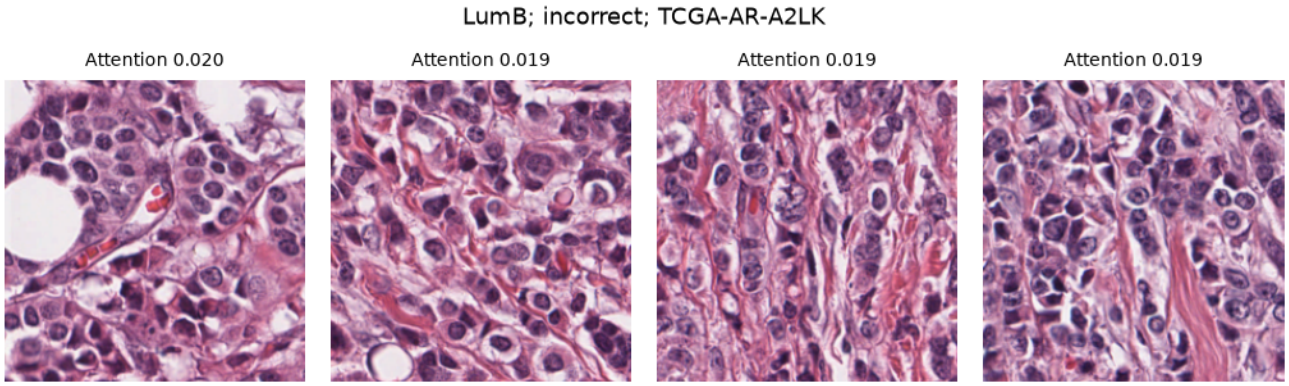
attention TCGA-AN-A0XU. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



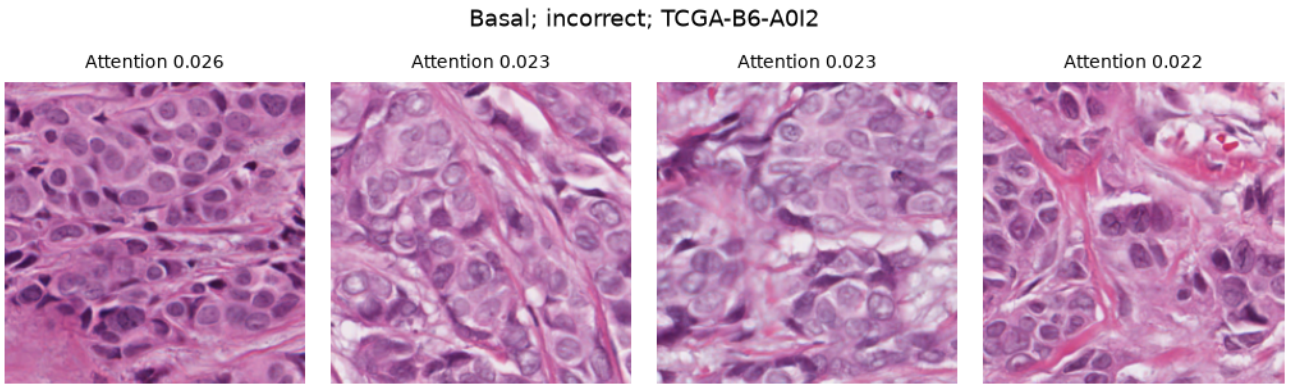
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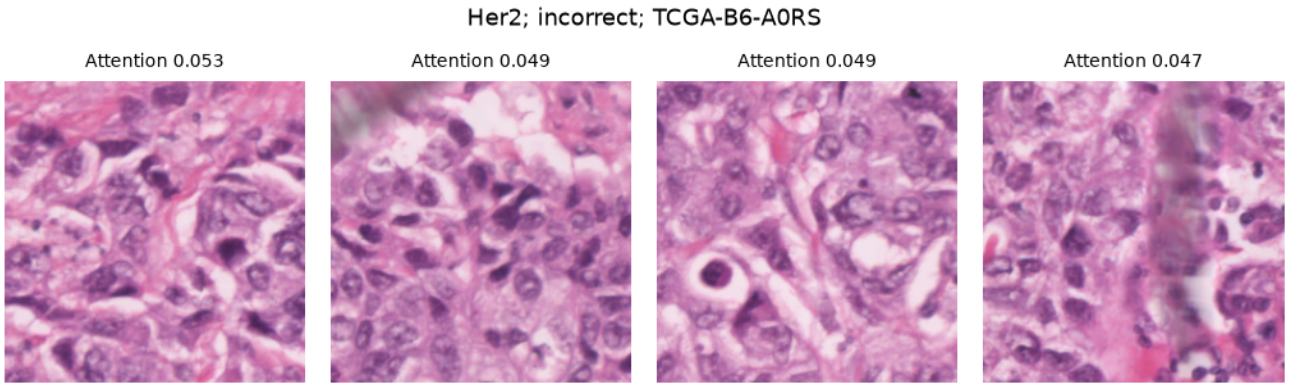
attention TCGA-AO-A1KQ. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



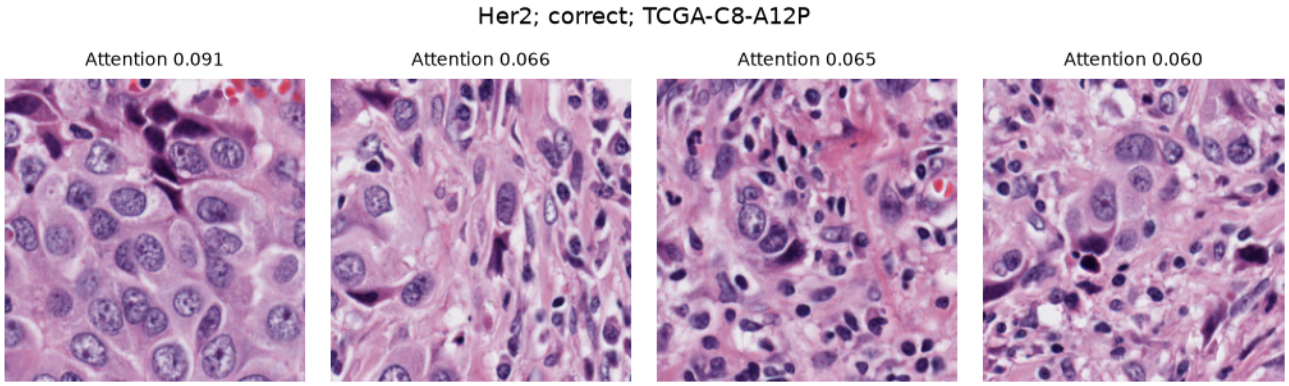
attention TCGA-AR-A2LK. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



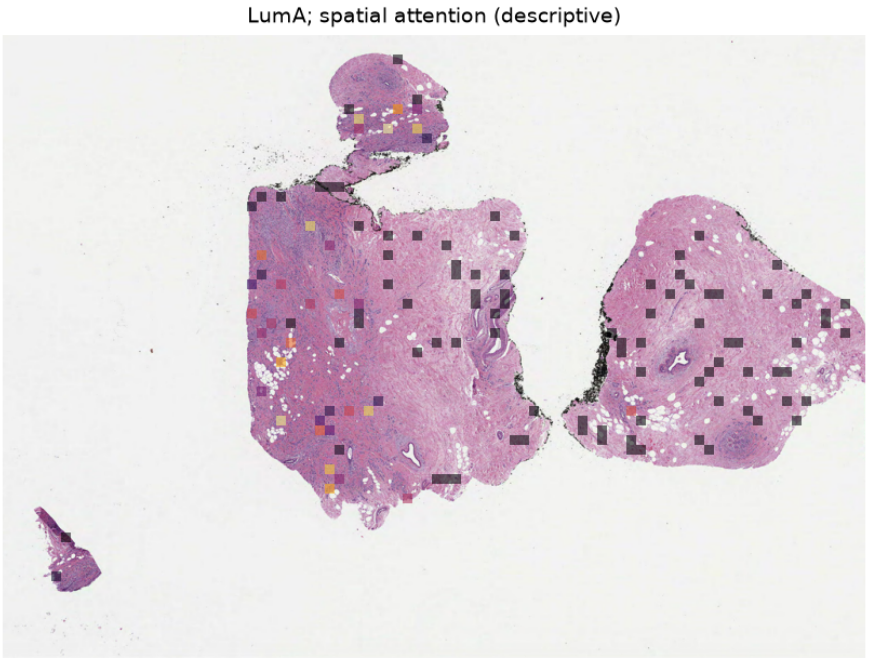
attention TCGA-B6-A0I2. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



attention TCGA-B6-A0RS. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.

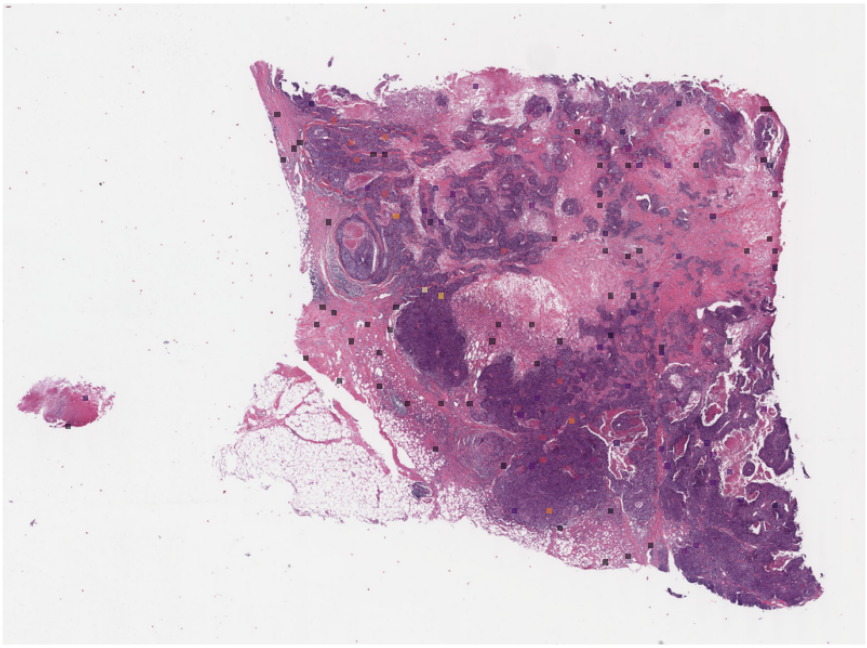


attention TCGA-C8-A12P. Top-attended tiles for a highest-confidence correct/incorrect case under the frozen logistic probe. The learned MIL weights describe image pooling, not validated localization.



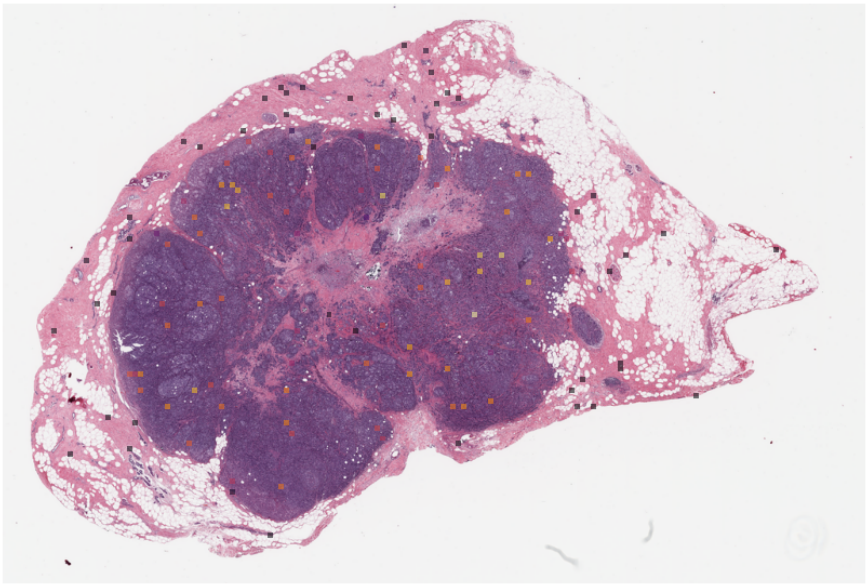
attention heatmap TCGA-AC-A2FG. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.

Basal; spatial attention (descriptive)

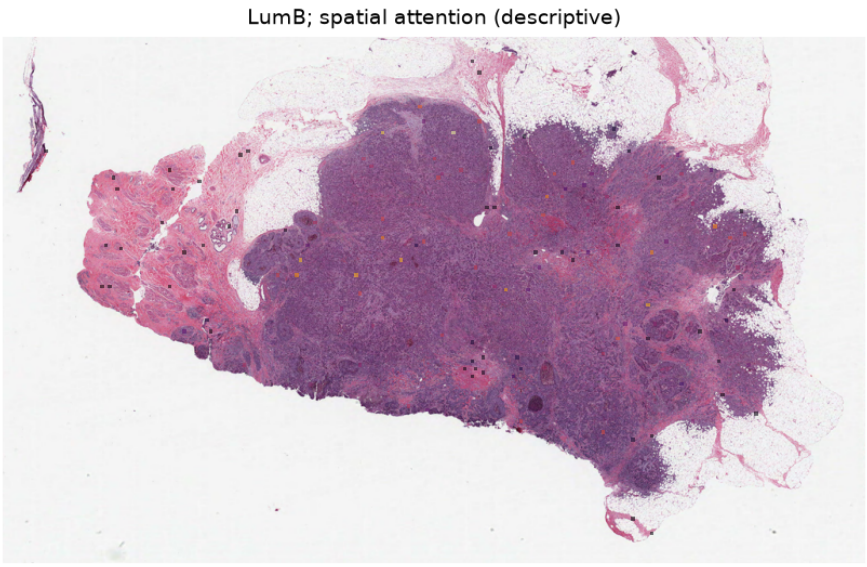


attention heatmap TCGA-AN-A0XU. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.

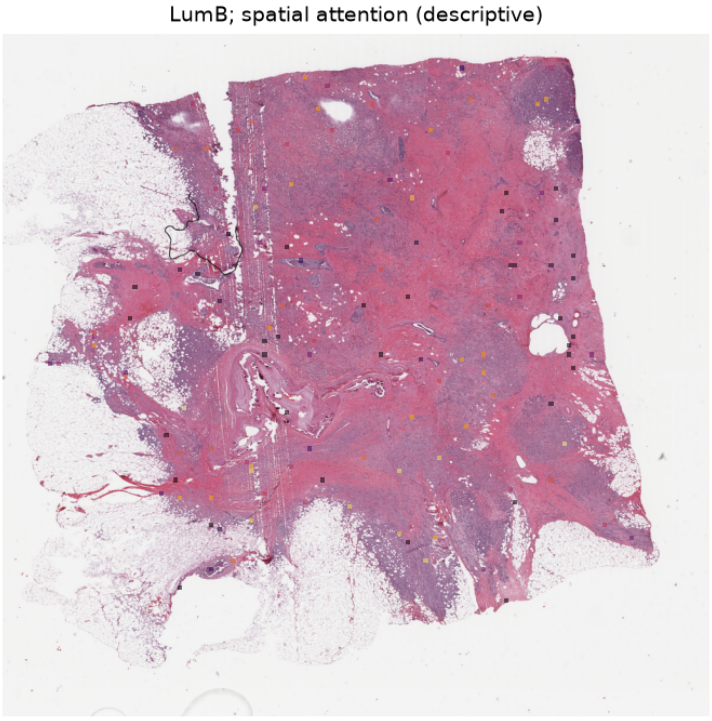
LumA; spatial attention (descriptive)



attention heatmap TCGA-AO-A0JI. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.

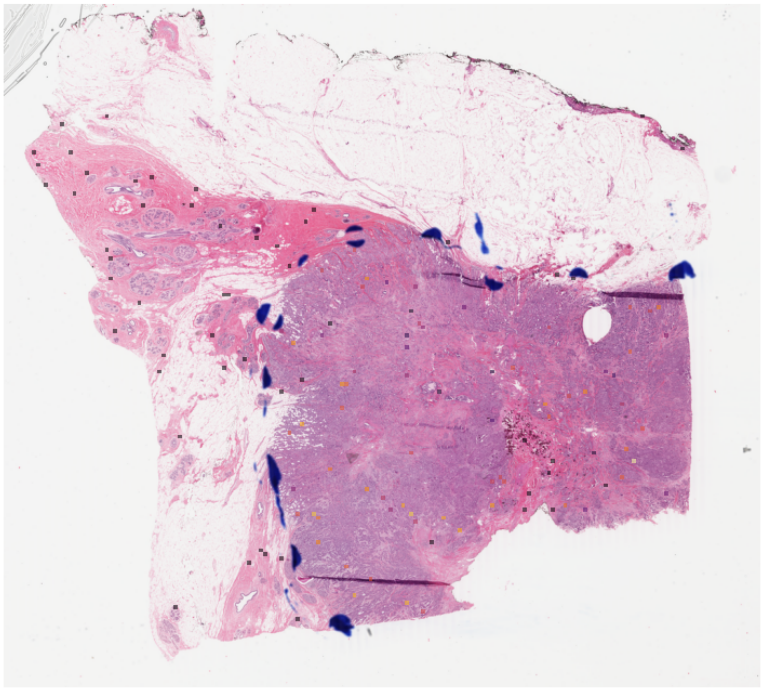


attention heatmap TCGA-AO-A1KQ. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.



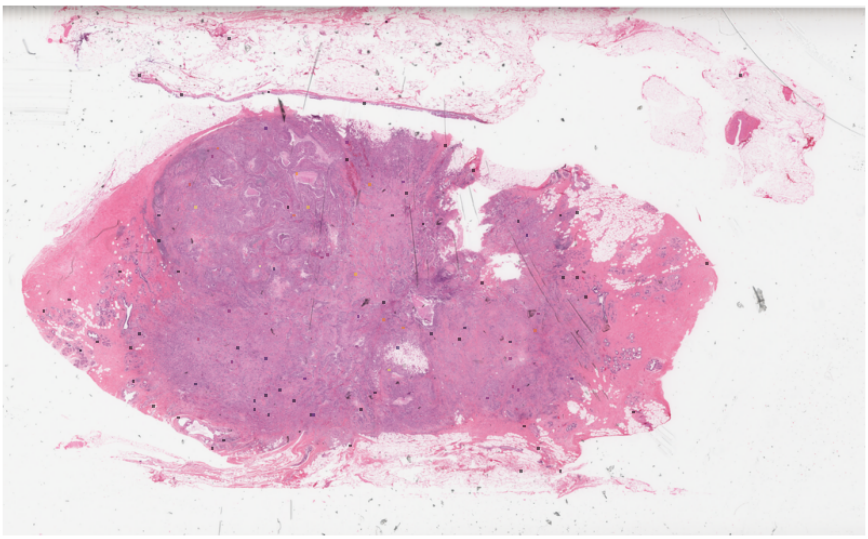
attention heatmap TCGA-AR-A2LK. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.

Basal; spatial attention (descriptive)

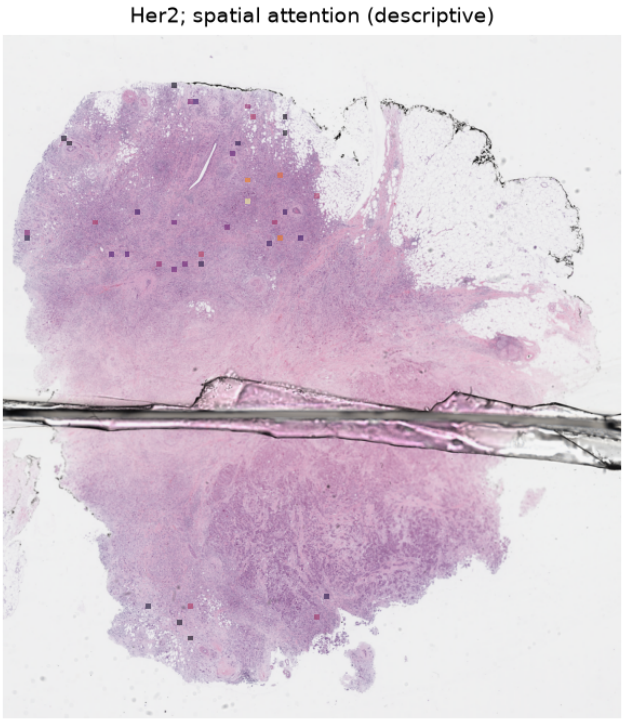


attention heatmap TCGA-B6-A0I2. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.

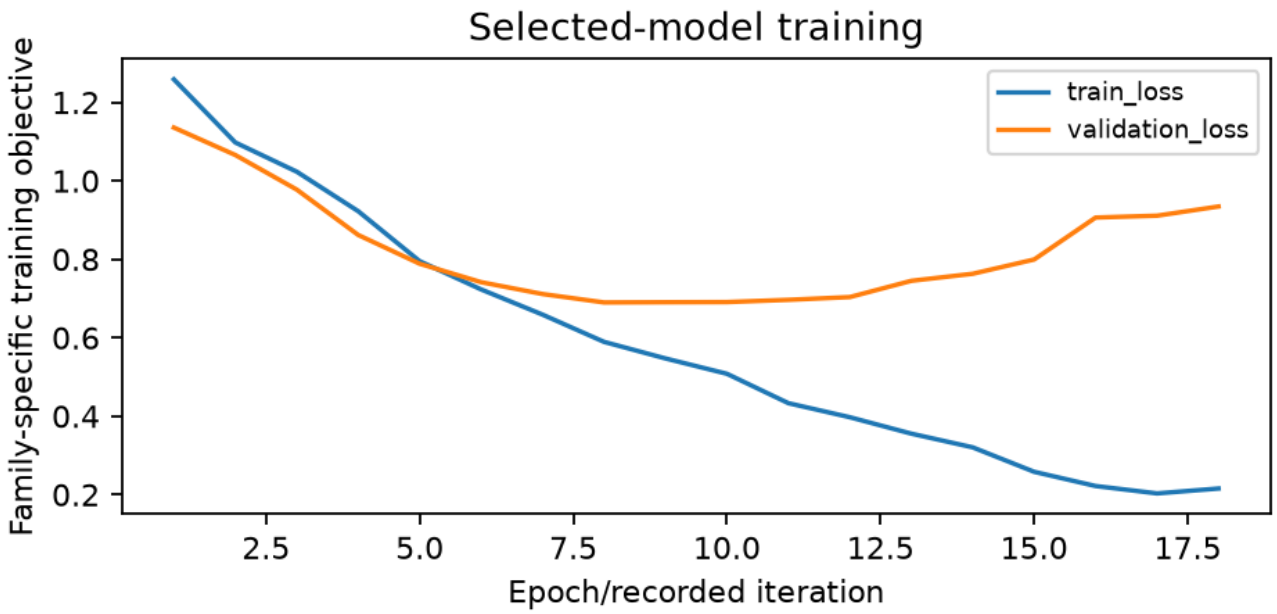
Her2; spatial attention (descriptive)



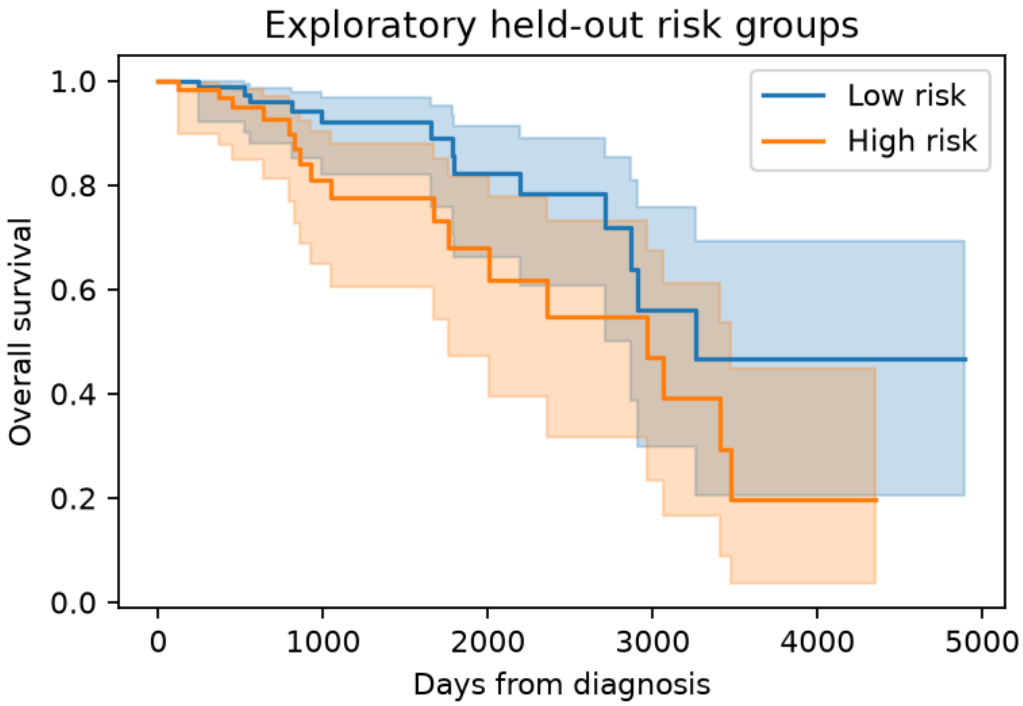
attention heatmap TCGA-B6-A0RS. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.



attention heatmap TCGA-C8-A12P. MIL attention on the highest-total-attention slide of the chosen patient; color is scaled to this slide's maximum weight. Cases follow the frozen logistic probe's confidence rule. This is descriptive pooling attention, not validated localization.



selected training curves. Training and validation objectives from the selected frozen run; selection used validation only.



survival km. Exploratory test-set Kaplan–Meier curves with 95% intervals. A penalized Cox model on training-fitted representation PCs defines risk; the training median risk sets the fixed high/low cutoff. This is retrospective stratification, not clinical validation.

Permitted interpretation, limitations and reproducibility

- Retrospective clinical data; missing covariates
- Histology may encode tissue source site
- PAM50 is an RNA-derived target; no RNA-derived input fields allowed
- Pathology FM pretraining overlap with TCGA cannot be excluded from local manifests
- Exploratory retrospective results do not establish prospective clinical benefit.
- Site-confounded embeddings do not support cross-site generalization claims.

Representations include sample IDs, split membership, model/checkpoint/preprocessing hashes and embedding definitions. The GPU ledger, environment locks, run manifests, decision log, split manifest and evidence source map accompany the outputs.

Pathology weights are local pinned resources licensed CC-BY-NC-ND-4.0 for academic/noncommercial use; neither those weights nor fine-tuned derivatives are redistributed. ImageNet features, where used, are labeled separately.

Sources: local data/label/model provenance manifests; pre-registered METHOD_NOTES.md; scvi-tools MultiVI, muon WNN, MOFA+, paired contrastive objectives and gated MIL implementations documented in the source and environment locks. Full citations are in the project report and docs/REFERENCES.json.